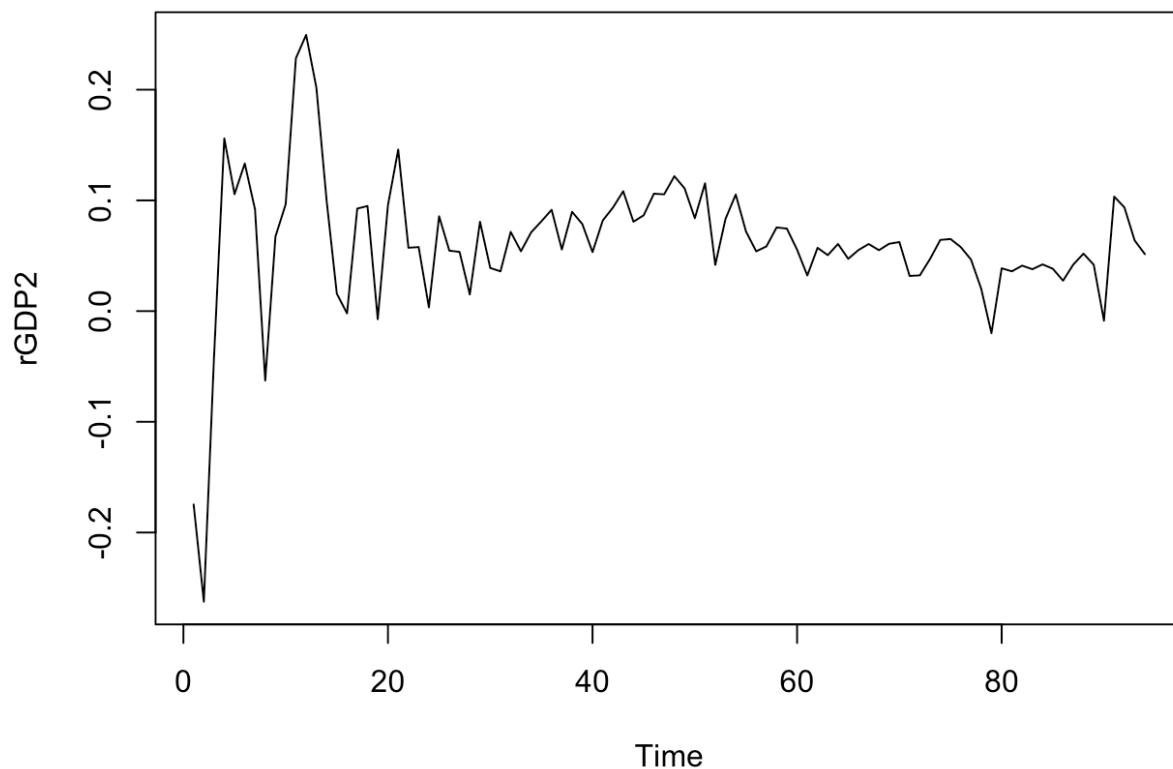


GDP Dataset

I pulled the GDP dataset for the US from 1930 to 2024 (www.bea.gov). I had to eventually move on from this dataset due to it being less than 100 points of data, which meant that it would require additional care as listed in the final project instructions. My goal was to test the rStudio model for a small dataset given that there was white noise in the Amazon dataset that I had pulled earlier for Matlab. I was trying to get an understanding of how to use rStudio while being able to run some of the models used in time series analysis such as GARCH and ARIMA models. This also includes a variety of tests such as the Adjusted Pearson Goodness-of-Fit Test, Sign Bias Test, Nyblom stability test, Weighted Ljung-Box Test on Standardized Squared Residuals and the Weighted Ljung-Box Test on Standardized Residuals.

I am including a few screens to show you the work that I did for the GDP dataset.

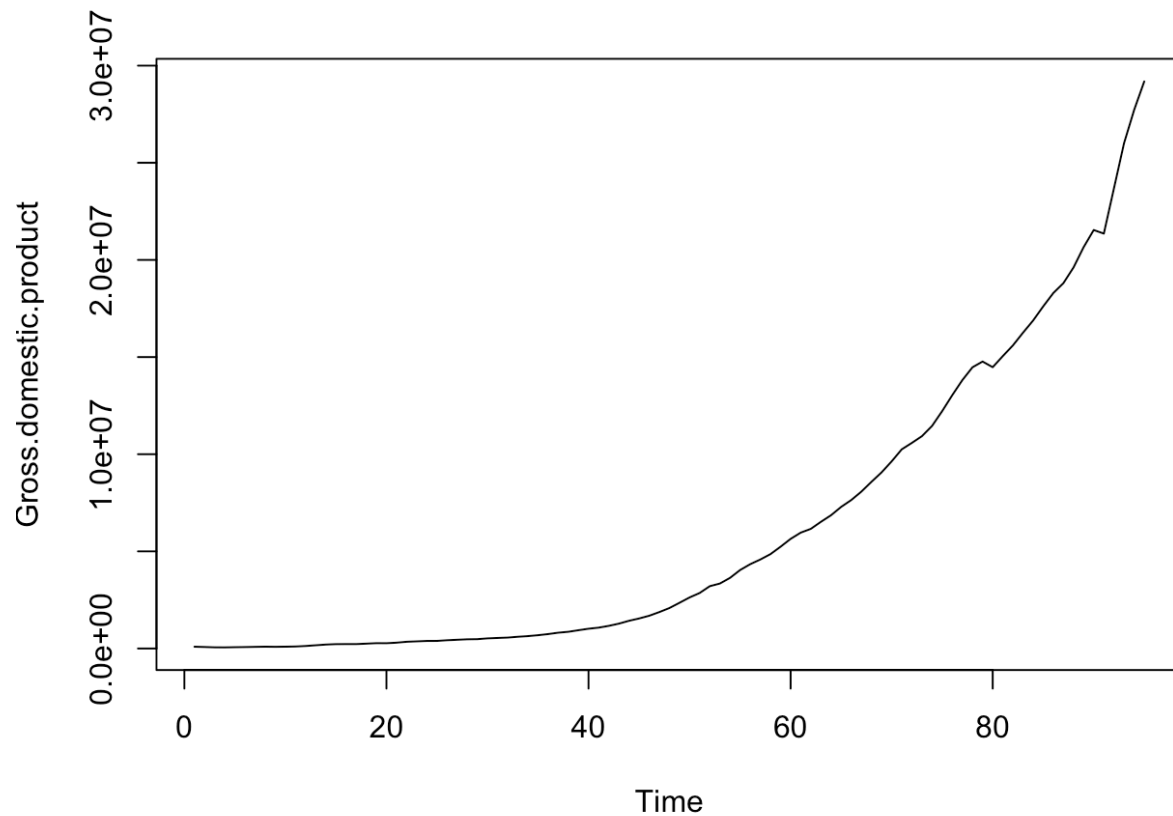


Remove stationary for GDP dataset:

```
> plot.ts(Gross.domestic.product)
> adf.test(Gross.domestic.product)
> plot.ts(rGD)
```

```
> attach(G2)
```

GDP Growth from 1930 to 2024:



Code:

```
rGDP2=diff(log(Gross.domestic.product))  
> adf.test(rGDP2)  
plot.ts(rGDP2)
```

ARIMA:

ARIMA(3,0,3) with non-zero mean

Coefficients:

	ar1	ar2	ar3	ma1	ma2	ma3	mean
	0.9741	-0.7386	-0.1618	-0.2979	0.3584	0.4239	0.0619
s.e.	0.3581	0.4190	0.3240	0.3414	0.2250	0.2016	0.0076

$\sigma^2 = 0.002264$: log likelihood = 155.27

AIC=-294.54 AICc=-292.84 BIC=-274.19

Exploratory Data Analysis:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
57154	389884	2081826	6139836	10416440	29184890

Gross.domestic.product	
nobs	9.500000e+01
NAs	0.000000e+00
Minimum	5.715400e+04
Maximum	2.918489e+07
1. Quartile	3.898835e+05
3. Quartile	1.041644e+07
Mean	6.139836e+06
Median	2.081826e+06
Sum	5.832844e+08
SE Mean	7.811825e+05
LCL Mean	4.588780e+06
UCL Mean	7.690892e+06
Variance	5.797337e+13
Stdev	7.614025e+06
Skewness	1.237608e+00
Kurtosis	4.882540e-01

ARCH-LM Test Results:

data: rGDP2

Chi-squared = 50.139, df = 12, p-value = 1.321e-06

Code:

```
> library(e1071)
> plot.ts(rGDP2)
> ArchTest(rGDP2)
```

GARCH Model Fit *

Conditional Variance Dynamics

GARCH Model : sGARCH(1,1)

Mean Model : ARFIMA(1,0,1)

Distribution : norm

Optimal Parameters

```
-----
      Estimate Std. Error t value Pr(>|t|)
mu    -0.099880   0.043658 -2.2878 0.022150
ar1    0.994481   0.006388 155.6688 0.000000
ma1   -0.602452   0.114880 -5.2442 0.000000
omega  0.000109   0.000052  2.0992 0.035803
alpha1 0.505498   0.203760  2.4809 0.013107
beta1  0.465550   0.125739  3.7025 0.000213
```

Robust Standard Errors:

```
      Estimate Std. Error t value Pr(>|t|)
mu    -0.099880   0.030683 -3.2552 0.001133
ar1    0.994481   0.005916 168.0983 0.000000
ma1   -0.602452   0.108469 -5.5541 0.000000
omega  0.000109   0.000053  2.0510 0.040267
alpha1 0.505498   0.209818  2.4092 0.015987
beta1  0.465550   0.121230  3.8402 0.000123
```

LogLikelihood : 182.601

Information Criteria

```
-----
Akaike      -3.7575
Bayes      -3.5951
Shibata    -3.7650
Hannan-Quinn -3.6919
```

Weighted Ljung-Box Test on Standardized Residuals

```
-----
              statistic p-value
Lag[1]          0.5445 0.4606
Lag[2*(p+q)+(p+q)-1][5]  4.2762 0.0318
Lag[4*(p+q)+(p+q)-1][9]  5.8866 0.2755
d.o.f=2
H0 : No serial correlation
```

Weighted ARCH LM Tests

Nyblom stability test

Individual Statistics:

Asymptotic Critical Values (10% 5% 1%)

Sign Bias Test

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	24.72	0.16985
2	30	34.94	0.20664
3	40	60.89	0.01398
4	50	60.26	0.13004

Elapsed time : 0.490967

```
> garch(rGDP2,grad="numerical",trace=FALSE)
```

Call:

```
garch(x = rGDP2, grad = "numerical", trace = FALSE)
```

Coefficient(s):

a0	a1	b1
0.0004715	0.3424809	0.5832913

Code for GARCH Test Results:

```
> rGDP2_garch_fit
> garch(rGDP2,grad="numerical",trace=FALSE)
```